



Cyber Investigation System

A secure, centralized platform for law enforcement to manage and investigate suspect data — built to simulate real-world tools like CCTNS/NCRP.

PYTHON · FLASK · MYSQL

The Problem



Traditional Systems Fall Short

Scattered Data

Information spread across multiple disconnected sources

Manual & Slow

Time-consuming searches with no automation

Weak Auth

Vulnerable authentication and high unauthorized access risk

Objectives



Secure Authentication

Session-based login to restrict unauthorized access



Suspect Data Management

Structured storage and retrieval of investigation records



Multi-Parameter Search

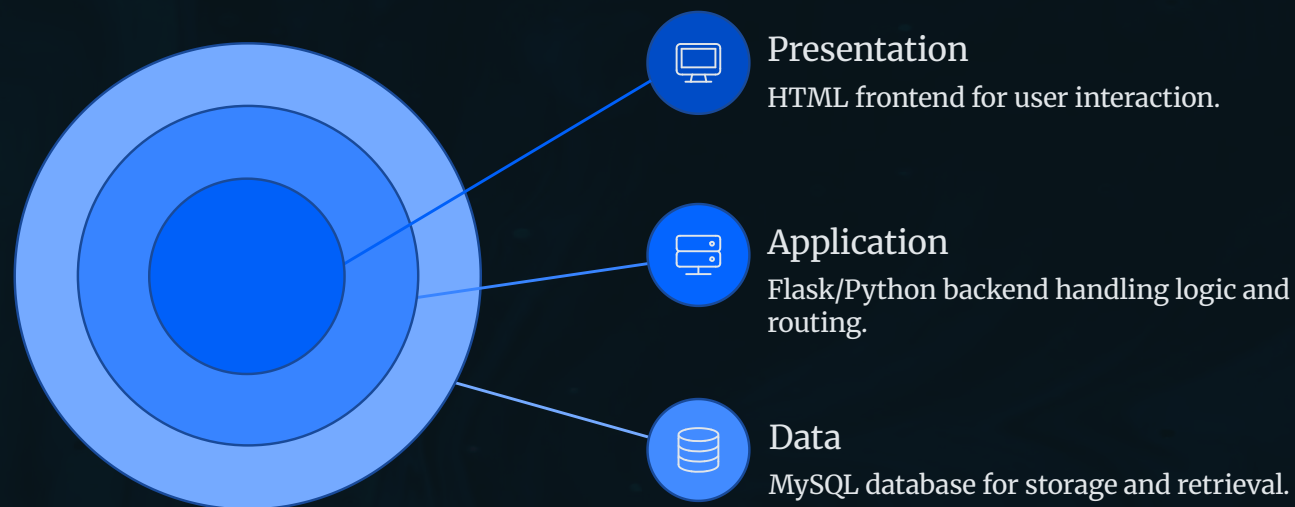
Search by name, mobile, Aadhaar, email, and more



Data Masking & Logs

Protect sensitive data and track all investigation activity

System Architecture



Data Flow

01

Login

User submits credentials; Flask verifies against MySQL

02

Dashboard

On success, central navigation interface is displayed

03

Query

User searches or adds data; query executes in database

04

Results

Masked results displayed; activity logged automatically

Technology Stack

Frontend

HTML, CSS

Backend

Python · Flask

Database

MySQL

Tools

VS Code · MySQL Workbench



Functional Modules

1

Authentication

Username/password login with session management

2

Dashboard

Central hub: Search, Add Record, View History, Logout

3

Data Entry

Add suspect records: name, mobile, email, Aadhaar, address

4

Search

Exact & partial match across 6 fields using SQL LIKE

5

Masking & Logs

Aadhaar shown as XXXX-XXXX-9012; all searches logged with timestamp

Database Design

Users Table

Field	Type
id	INT (PK)
username	VARCHAR
password	VARCHAR

Search Logs Table

Field	Type
id	INT
username	VARCHAR
keyword	VARCHAR
search_time	TIMESTAMP

Records Table

Field	Type
id	INT
name	VARCHAR
father_name	VARCHAR
mobile	VARCHAR
email	VARCHAR
Aadhar	VARCHAR
address	TEXT

Security & Testing

Security Features



Session Auth



Route Protection



Data Masking



Input Validation

Test Cases

Scenario	Result
Valid login	✓ Access granted
Invalid login	✗ Error message
Add record	✓ Data stored
Search valid data	✓ Results shown
Search invalid data	⚠ No result

Advantages, Limitations & Future

✓ Strengths

- Fast, efficient multi-field search
- Secure access control & session management
- Real-world simulation of investigation tools
- Scalable 3-tier architecture

⚠ Limitations

- Basic UI; no encryption or API integration
- Not deployed online

🚀 Future Enhancements





Conclusion

The Cyber Investigation System is a functional prototype that successfully integrates **secure authentication, data management, smart search, data masking, and activity tracking** — forming a strong foundation for advanced cyber investigation platforms.

Developer

Vicky

Role

Full Stack Developer

Stack

Python · Flask · MySQL